Chau Pham

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Education

University at Buffalo Aug 2023 - now

- Ph.D Student in Computer Science and Engineering. Advisor: David Doermann.

Hanoi University of Science and Technology Sep 2017 - May 2022

- B.Sc, Mechatronics, Honor program.

Research and Work Experience

University at Buffalo - Research Assistant Aug 2023 - now

Research topic:

- Large Vision-Language Model and AI agent applications for scene understanding and image generation tasks.
- RL post-training for Image/Video generation and Image/Video editing.

Research projects:

- Proposed an RL-based post-training framework for adapting pretrained text-to-image models to instruction-based image editing without paired image data, achieving competitive performance with supervised fine-tuning under the same setting [pdf].
- Proposed an RL-based hyperparameter optimization framework for Prompt-to-Prompt-style image editing, enabling efficient discovery of near-optimal editing configurations [pdf].
- Proposed a test-time training-free personalization method for vision-language models (VLMs), enabling personalized predictions without model fine-tuning [pdf].

VinAI Research *(now Qualcomm AI Vietnam)* - AI Research Resident Dec 2020 - July 2023

Research topics: Weakly supervised learning for perception algorithms, including segmentation and object detection.

Research projects:

- Developed an open-vocabulary object detection framework that distills knowledge from pretrained CLIP and retrieves relevant object concepts at test time, improving generalization to unseen categories. [pdf]
- Introduced a benchmark dataset and learning framework for joint object counting and detection.[pdf]
- Proposed a weakly supervised 3D object detection method using 2D bounding-box annotations and developed a tool for evaluating the annotation cost associated with 3D object detection.

Publications and Preprint

- **Chau Pham**, Quan Dao, Vishnu Lokhande and David Doermann "ControlEdit: Image Editing with Unpaired Post-Training", under review.[pdf]
- **Chau Pham**, Quan Dao, Mahesh Bhosale, Yunjie Tian, Dimitris Metaxas, and David Doermann "AutoEdit: Automatic Hyperparameter Tuning for Image Editing", in NeurIPS 2025.[pdf]

- **Chau Pham**, Hoang Phan, David Doermann, and Yunjie Tian, "Personalized Large Vision-Language Model", in MMFM Workshop, CVPRW 2025. [pdf]
- **Chau Pham**, Tuan Truong Vu, and Khoi Nguyen, "LP-OVOD: Open-Vocabulary Object Detection by Linear Probing on Pseudo Labels Retrieved from Top Relevant Box Proposals", in *Winter Conference on Applications of Computer Vision (WACV)*, 2024. [pdf]
- Thanh Nguyen*, **Chau Pham***, Khoi Nguyen, and Minh Hoai, "Few-Shot Object Counting and Detection", in *European Conference on Computer Vision (ECCV)*, 2022. [pdf]
- Hoang Phan, Minh Pham, **Chau Pham**, Chinmay Hegde, Trung Le, Qi Lei, "Rationale-Guided Policy Optimization: Learning to Reason with Adaptive Rationale Scaffolding", in NeurIPS 2026.
- Mahesh Bhosale, Naresh Kumar Devulapally, Abdul Wasi, **Chau Pham**, Vishnu Suresh Lokhande, and David Doermann, "Variance-Guided Score Regularization for Hallucination Mitigation in Diffusion Models", under review.

(∗) denotes equal contribution.

Technical skills

- **Programming Language and Framework**: Python, C++, PyTorch, HuggingFace (Transformers, Diffusers, Accelerate, PEFT) and OpenCV.
- **Generative Model**: Diffusion/Flow Matching model, Large Language Model/Vision-Language Model, Text-to-image/Video model, CLIP.
- **Modern Deep Learning techniques**: Model training, LoRA fine-tuning and Reinforcement Learning post-training.

Professional services

- Reviewer at ICLR, CVPR, ICCV, NeurIPS.
- TA at UB (CSE 191, CSE 674, CSE 473/573).